

Homework Assignment: Classifier Evaluation

1 Learning Objectives

Upon completing this assignment, you should be able to:

- Calculate and interpret common metrics for evaluating binary classifiers.
- Explain the effects of class imbalance and classification thresholds on classifier evaluation.
- Construct and interpret ROC and Precision–Recall curves.
- Apply cross-validation to select hyperparameters while maintaining an independent test dataset.
- Build a leakage-resistant machine-learning pipeline for regularized logistic regression.
- Evaluate and interpret the generalization performance of a logistic regression classifier using biological data.

2 Solutions

2.1 Problem 1: Conceptual Understanding of Classifier Evaluation

2.1.1 Performance Metrics

From the confusion matrix,

$$TN = 820, \quad FP = 80, \quad FN = 30, \quad TP = 70.$$

Thus,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 820}{1000} = 0.89 = 89\%.$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{70}{100} = 0.70 = 70\%.$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{820}{900} \approx 0.911 = 91.1\%.$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{70}{150} \approx 0.467 = 46.7\%.$$

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} = \frac{140}{250} = 0.56.$$

2.1.2 Accuracy and Class Imbalance

Accuracy alone is not adequate. Only 100 of 1,000 patients (10%) have the disease. A classifier that predicts “no disease” for every patient would achieve 90% accuracy, yet its sensitivity would be zero because it would miss every patient with disease. Therefore, accuracy should be considered together with sensitivity/recall, specificity, precision, and F1 score, as well as the consequences of false negatives and false positives.

2.1.3 Effect of Changing the Classification Threshold

Decreasing the threshold from 0.50 to 0.30 makes positive classification easier. In general,

$$TP \uparrow, \quad FN \downarrow, \quad FP \uparrow, \quad TN \downarrow,$$

or each quantity may remain unchanged if no predicted probabilities lie between the two thresholds. Consequently, recall generally increases or remains unchanged. Precision often decreases because additional positive predictions may include more false positives, although precision is not guaranteed to decrease monotonically at every threshold.

Lowering the threshold tends to identify more actual positive cases, improving recall, but it can also classify more negative cases as positive. This produces the precision–recall tradeoff.

2.1.4 ROC versus Precision–Recall Curves

An ROC curve plots true-positive rate (sensitivity) against false-positive rate (1–specificity) as the classification threshold changes. A PR curve plots precision against recall as the threshold changes. Particular attention should be paid to the PR curve here because disease is the minority class (10% prevalence). Precision directly reflects false-positive predictions among predicted positives and depends on positive-class prevalence. The PR curve is therefore especially informative when the positive class is rare and identifying positive cases is of primary interest. The ROC curve remains useful and provides complementary information about discrimination across both classes.

2.1.5 Hyperparameter Tuning and the Test Dataset

The proposed procedure uses the test set to select C . The test data would therefore influence model development, and the resulting test AUC would no longer be an independent estimate of generalization performance. A correct workflow is

$$\boxed{\text{All Data}} \longrightarrow \boxed{\text{Development Data} + \text{Test Data}}.$$

Set the test data aside. Using only development data,

$$\boxed{\text{Development Data}} \longrightarrow \boxed{\text{Cross-Validation}} \longrightarrow \boxed{\text{Select } C}.$$

After selecting C , fit the final model using all development data:

$$\boxed{\text{Selected } C} \longrightarrow \boxed{\text{Train Final Model on All Development Data}}.$$

Finally,

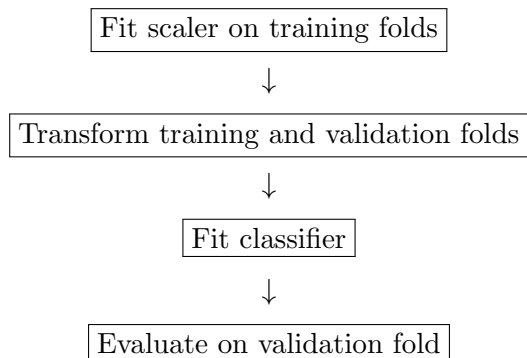
$$\boxed{\text{Final Model} + \text{Untouched Test Data}} \longrightarrow \boxed{\text{Final Performance Estimate}}.$$

The test set should be used only for final evaluation, not for selecting C , features, preprocessing procedures, or other modeling decisions.

2.1.6 Feature Scaling and Information Leakage

Scaling should occur inside cross-validation. In each fold, the scaler’s parameters (e.g., mean and standard deviation) are estimated using only the training folds and then applied to the validation fold. Standardizing all development data before cross-validation would allow validation samples to

influence preprocessing and would therefore cause information leakage. A pipeline implements the appropriate sequence:



Scaling is especially important with regularization because the penalty depends on coefficient magnitude. For L2 regularization,

$$\lambda \sum_{j=1}^d \theta_j^2$$

penalizes large coefficients. Predictors measured on very different scales would otherwise be affected differently by the penalty. Standardization places continuous predictors on comparable scales.